Zhan Yang

Research Interest: Railway Track Inspection and Monitoring, Ground-Penetrating Radar, Machine Learning

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Nationality:	Chinese	Date of birth:	2000.12.23

Education

Southwest Jiaotong University—Chengdu, China **09/2025-**

- Doctoral Student of Road and Railway Engineering, School of Civil Engineering
- Supervisor: Prof. Rongshan Yang

Southwest Jiaotong University—Chengdu, China **09/2022-06/2025**

- Master Student of Road and Railway Engineering, School of Civil Engineering **|** Average Score:88.22/100
- Supervisor: Prof. Rongshan Yang

Southwest Jiaotong University—Chengdu, China **09/2018-06/2022**

- Bachelor of Railway Engineering, School of Civil Engineering
- Average Score (Core courses): 88.00/100 **|** Rank: 5/145 (Top 3.5%)

Research Experience

Evaluating Railway Ballasted Bed Conditions Using Ground-Penetrating Radar (GPR) **09/2023-Now**

Research Project, Principal Investigator

- Conducted a comprehensive literature review to understand the current state of research and development trends in this field.
- Investigated railway sleeper interference characteristics on GPR signals, while developing an unsupervised learning-based algorithm to suppress this interference.
- Developed an automated method for predicting ballast layer thickness using graph neural networks.
- Designed a calibration test line to investigate the characteristic signal features associated with ballast fouling and ballast bed defects.

In-Service Condition Testing and Evaluation of Ballasted Track Before and After Overhaul Operations – Field Experiment on an Operating Railway Line **05/2025**

Experimental study project, Principal Investigator

- Involved in the full-cycle execution of a field study on operating railway lines, including experimental design, task coordination, material preparation, logistics management, pre-test validation, and coordination with the railway bureau, station departments, and field personnel.
- Organized an 8-member team in performing ballast longitudinal and lateral resistance testing, infrared thermography, and ground-penetrating radar (GPR) surveys.
- Gained hands-on experience in planning and executing large-scale field experiments involving multi-party collaboration and complex operational conditions.

State Evolution and Performance Regulation of Continuously Welded Rail at the Bridge End of Long-Span Bridges on High-Speed Railways under Complex Operating Conditions **02/2025-04/2025**

NSFC–China Railway Joint Fund (Proposal Stage), Core member

- Actively involved in the entire proposal development process, including project conceptualization, proposal drafting, and refinement.
- Gained comprehensive experience in national joint research fund applications, with a good working knowledge of proposal writing standards, procedural workflows, and team collaboration practices.

Publication

- **Z. YANG**, R. YANG, X. LIANG, et al. Automatic Prediction of Railway Ballast Layer Thickness Using Graph Neural Network Based on GPR Reflection Signals [J]. Automation in Construction, 2025, 177: 106318. **Online**
- **Z. YANG**, R. YANG, C. LAN, et al. SN-1DcycleGAN: An Unsupervised Learning-Based Network for Sleeper Interference Suppression in Ballasted Track GPR Data [J]. IEEE Transactions on Geoscience and Remote Sensing, 2024, 1-14. **Online**
- C. LAN, **Z. YANG**, X. LIANG, et al. Experimental Study on Wayside Monitoring Method of Train Dynamic Load Based on Strain of Ballastless Track Slab [J]. Construction and Building Materials, 2023, 394: 132084. **Online**
- X. LIANG, R. YANG, H. QIAN, **Z. YANG**, et al. Observing and identifying fouled ballast bed using infrared thermography (IRT): A real-time temperature prediction study based on an enhanced BiGRU model[J]. Mechanical Systems and Signal Processing, 2025, 224: 112150. **Online**

Awards and Honors

- Outstanding Graduates in Sichuan Province, China **04/2025**
- Merit Graduate Student **12/2023,12/2024**
- First-Class Entry Scholarship; First-Class Academic Scholarship **10/2022; 10/2023,10/2024**
- Outstanding Undergraduates in Sichuan Province, China **03/2022**
- First-Class Comprehensive Scholarship **12/2020**
- Merit Undergraduate Student **12/2020, 12/2021**

Skills

Languages:

Chinese: Mother Tongue English: CET-6(Chinese College English Test), IELTS(Preparing)

Computer skills:

Python (Pytorch, Tensorflow), Remote Development, GprMax, Solidworks, Zotero, Origin, Microsoft Office